Masum - 50

CH - 5 Reaction Mechanism of d-block complex

Q: Explain the stability & lability for complex compound.

Thermodynamic stability and libility have no direct relations. A compound which is thermodynamically stable may be labile (অস্থির)

Ex:

$$[Ni(CN)_4]^{2-} + 4\,{}^{14}CN^- \rightleftharpoons [Ni({}^{14}CN)_4]^{2-} + 4CN^-$$

On the other hand, $[Co(NH_3)_6]^{3+}$ is not stable but it is a inert (নিষ্ক্রিয়). In acid soln, it

$$2[Co(NH_3)_6]^{3+} + 10H^+ + 18H_2O \longrightarrow 2[Co(H_2O)_6]^{2+} + 12NH_4^+ + 1/2\,O_2$$

(Very slow)

F-22 F-21 → b(b)

$$[Co(NH_3)_5(CO_3)]^{+} + 2H_3O^+ \longrightarrow [Co(NH_3)_5(H_2O)]^{3+} + CO_2 + 2H_2O$$

Mechanism:

$$[(NH_3)_5 Co^{(III)} - \ddot{O} - CO_2]^{+} + H_3O^+$$

$$\downarrow$$

$$[(NH_3)_5 Co^{(III)} - \ddot{O}(H^+)(CO_2)]^{2+} + H_2O$$

$$\downarrow$$

$$[(NH_3)_5 Co^{(III)} \leftarrow \ddot{O}(H)(H)]^{3+} + H_2O$$

So, in this complex, "O" of H_2O is from CO_3^{2-} and is not replaced CO_3^{2-} by H_2O

বা

$$[{}^{56}Fe^{3+}(CN)_6]^{3-} + [{}^{59}Fe^{2+}(CN)_6]^{4-} \longrightarrow$$

$$[{}^{59}Fe(CN)_6]^{3-} + [{}^{56}Fe(CN)_6]^{4-}$$

This reaction is not substitution, it is a redox rxn. Actually, e^- transfer from $[{}^{59}Fe(CN)_6]^{4-}$ to $[{}^{56}Fe(CN)_6]^{3-}$

$${}^{59}Fe^{2+} + {}^{56}Fe^{3+} \longrightarrow {}^{59}Fe^{3+} + {}^{56}Fe^{2+}$$

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Mechanism of ligand substitution:

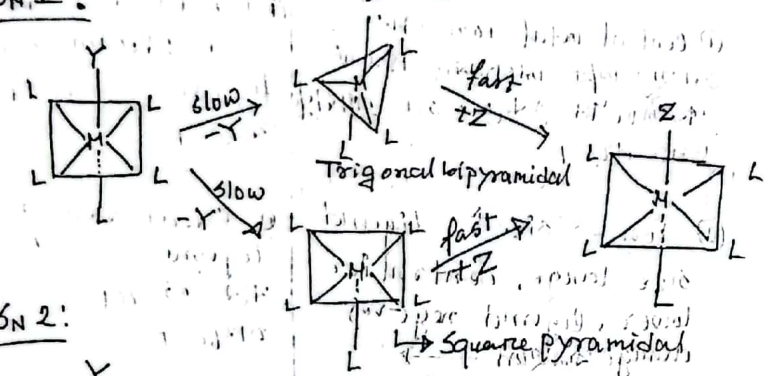
① S_N1 : (dissociative) & (ii) S_N2 (Associative)

Difinition

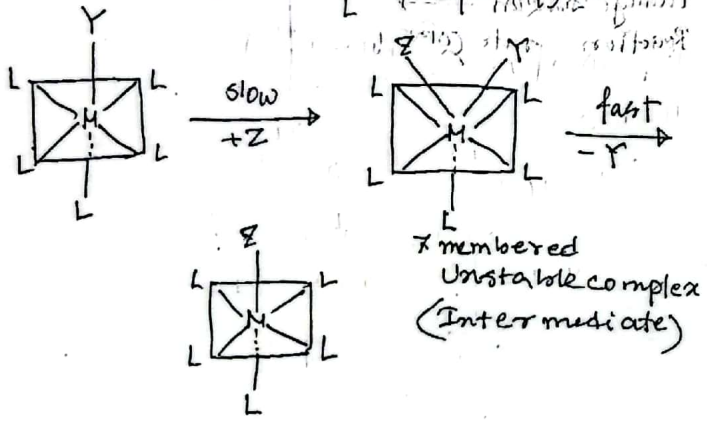
F-20

$$[L_5MY] + Z \longrightarrow [L_5MZ] + Y$$

F-19 S_N1 :



F-20 S_N2 :



Usually → Octahedral → S_N1

Tetra/square planer → S_N2

S_N1

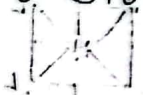
① Central metal ion এবং লিগ্যান্ডের মধ্যে Nature এর উপর নির্ভর করে। কোঅর্ডিনেশন সংখ্যা উপর নয়।

② Ion → size ↑, ligand size large, central ion lower, ligand neg (-ve) charge বেশি হলে → Reaction rate কমে।



S_N2

① Central ion এবং আক্রমণকারী লিগ্যান্ডের মধ্যে ligand-এ জোর নির্ভর করে।



② Central metal ion & ligand এর মধ্যে bond বন্ধ এর উপর rate নির্ভর।

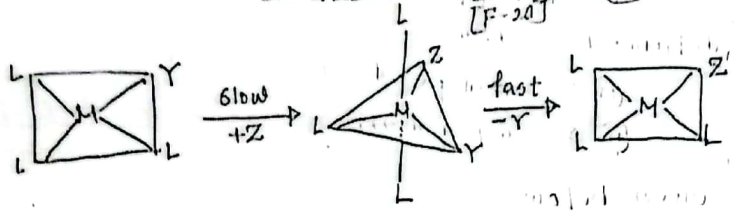




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Substitution of square planar Complex:

Dissociative Mechanism (CE-2) → (2)
[E-2]



2nd Order

$$ML_3Y + Z \longrightarrow ML_3Z + Y$$

$$\text{rate} = k[ML_3Y]^1[Z]^1$$

$$\text{Order} = 1+1 = 2$$

Water & other solvent may affect the rate of reaction. Because they can act as a ligand. So, the rate will be —

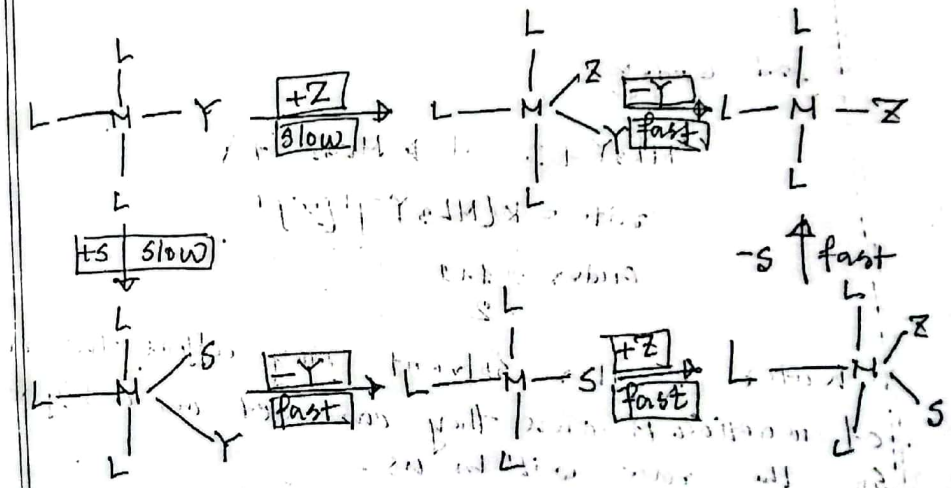
$$-\frac{d}{dt}[ML_3Y] = K_1[ML_3Y] + K_2[ML_3Y][Z]$$

K_1 & K_2 rate constant of 1st order & 2nd order reaction.

From the two derived rate law, it is proved that, the reaction will proceed in 2 different paths.

(i) Z independent
(ii) Z dependent

Given Below:



Z Independent, S participate করে।

$$K_1 = K[S]$$

(H_2O, Alcohol)

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Z dependent. 2nd order, I_dS_N2 mechanism.

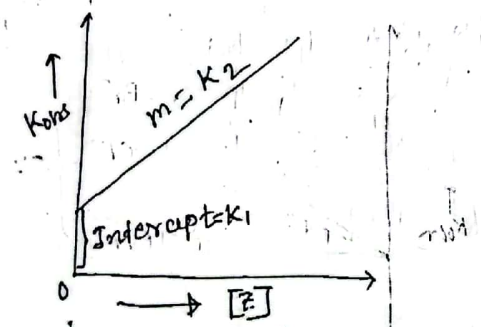
In solvent, CCl_4, C_6H_6, C_6H_{14} etc.

If the amount of ligands is very high then the reaction will be pseudo-first-order.

In this case,

$$-\frac{d}{dt}[ML_3Y] = K_1[ML_3Y] + K_2[ML_3Y][Z]$$
$$= \{K_1 + K_2[Z]\} \times [ML_3Y]$$
$$= K_{obs} \times [ML_3Y]$$
$$\therefore K_{obs} = K_1 + K_2[Z]$$

Plotting K_{obs} vs $[Z]$



C_6H_{14}, CCl_4, C_6H_6 etc solvent ए $K_1 = 0$ अर्थात्,

$K_{obs} = K_2[Z]$

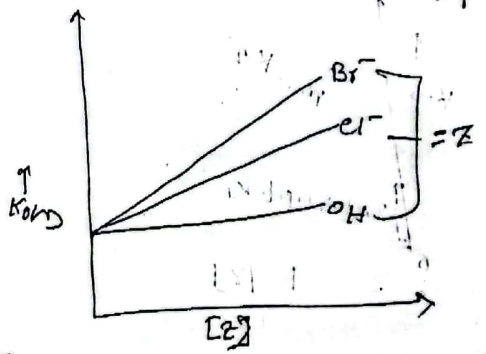
(ligand dependent)

In H_2O, C_2H_5OH etc प्रकार solvent path & ligand path का कोई competition होता.

In a fixed solvent, different ligands react with complexes and getting K_{obs} vs $[Z]$ are straight line. These ~~intercept~~ intersect the same point but the slopes are different.

K_1 constant but K_2 different

↳ बदले ligand different



$[Pt(dien)Cl]^+ + Z \rightarrow Pt[(en)Z]^{2+} + Cl$

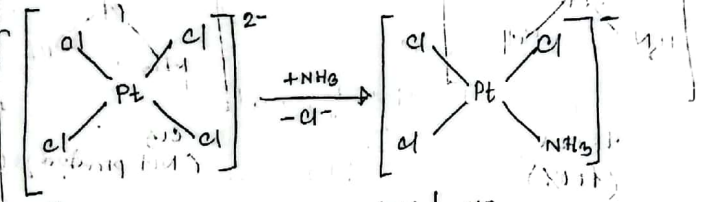
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Nucleophilicity sequence.

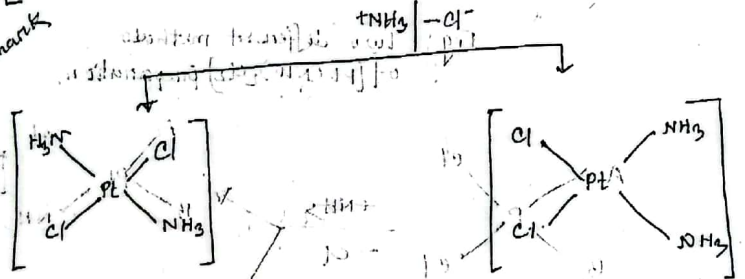
$F^- < H_2O < NH_3 \sim Cl^- < Br^- < Py < I^- < CN^-$

Trans effect :→ Definition → F-22 F-21

Trans effect is the increase of lability of ligands that are trans to certain other ligands in the substitution reaction of square planar complex.



F-20 (revised) 4 mark



Trans isomer
(Not produce yet)

cis-isomer
(100%)

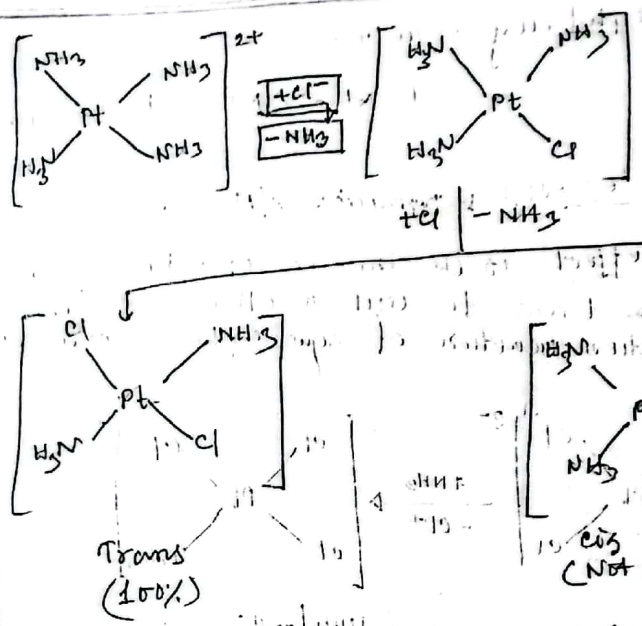
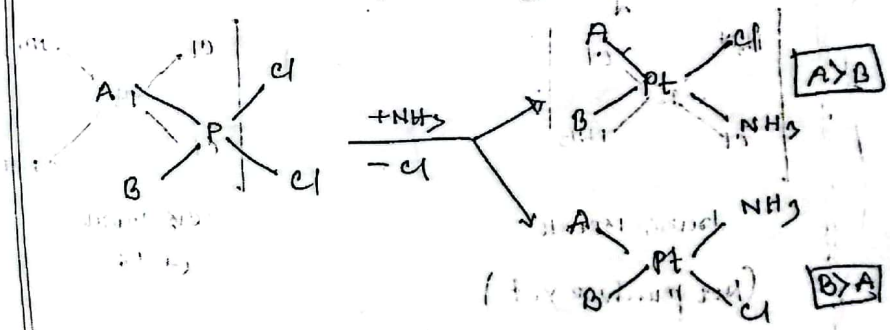


Fig: Two different methods of $[Pt(NH_3)_2Cl_2]$ preparation.



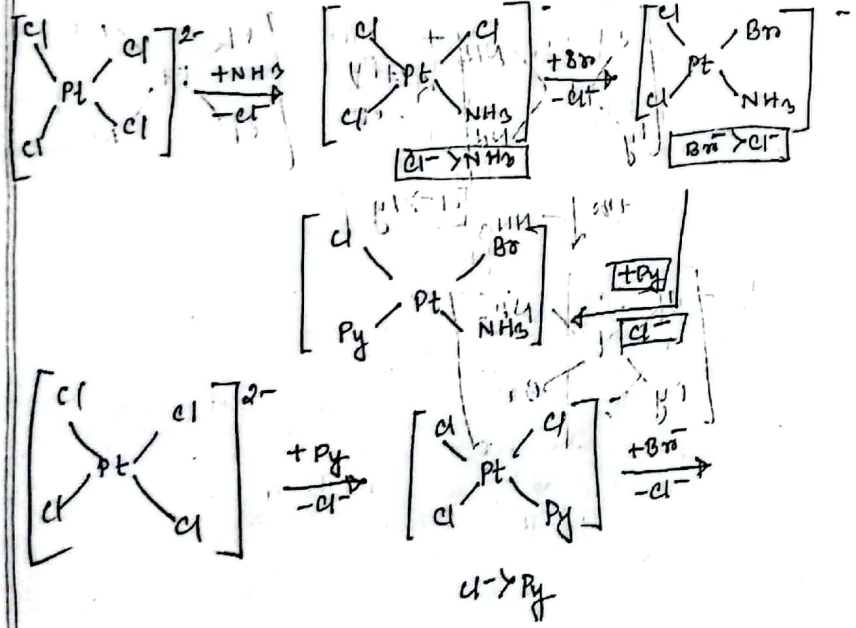
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Sequence of trans effect power :

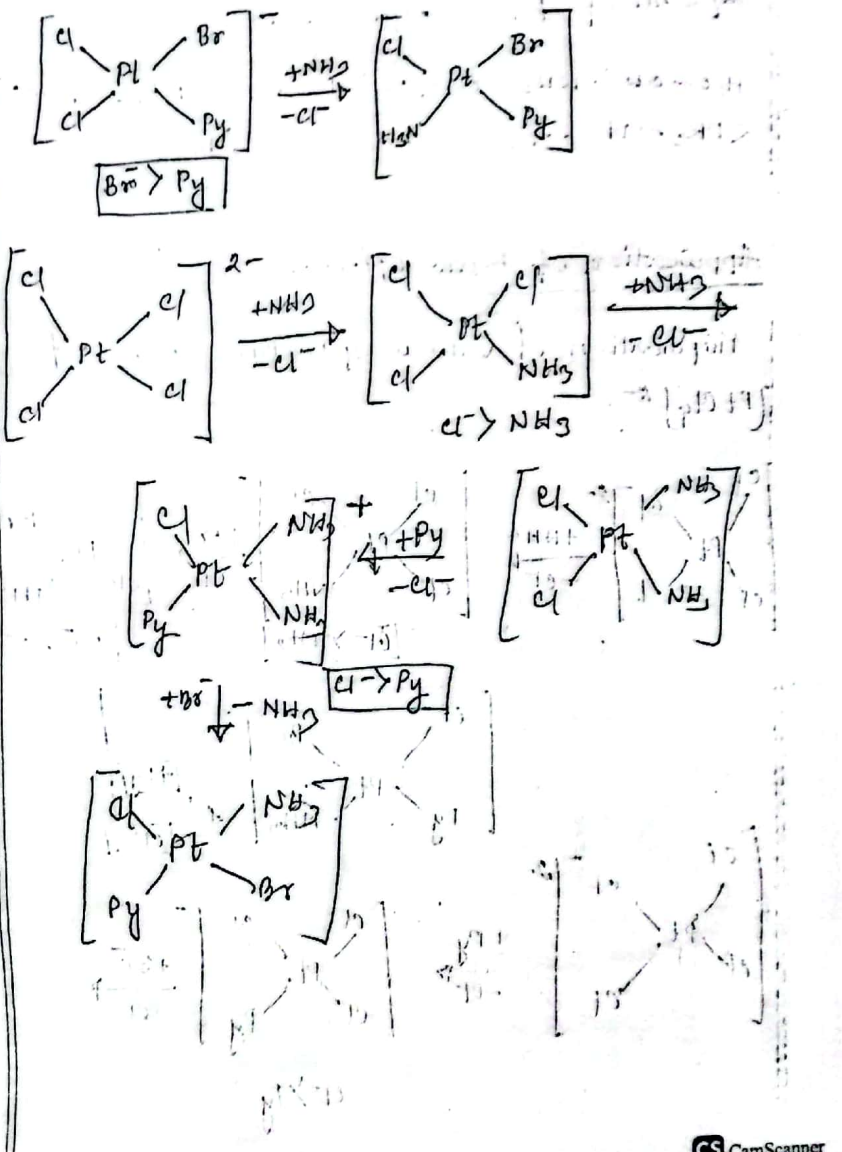
$H_2O \sim OH^- \sim NH_3 \sim Py < Cl^- < Br^- < I^- \sim NO_2^- < C_6H_5^- < CH_3^-$
$< PR_3 \sim H^- << NO \sim CO \sim CN^- \sim C_2H_4$

Application of trans effect :

Preparation of various geometrical isomers from the $[PtCl_4]^{2-}$.



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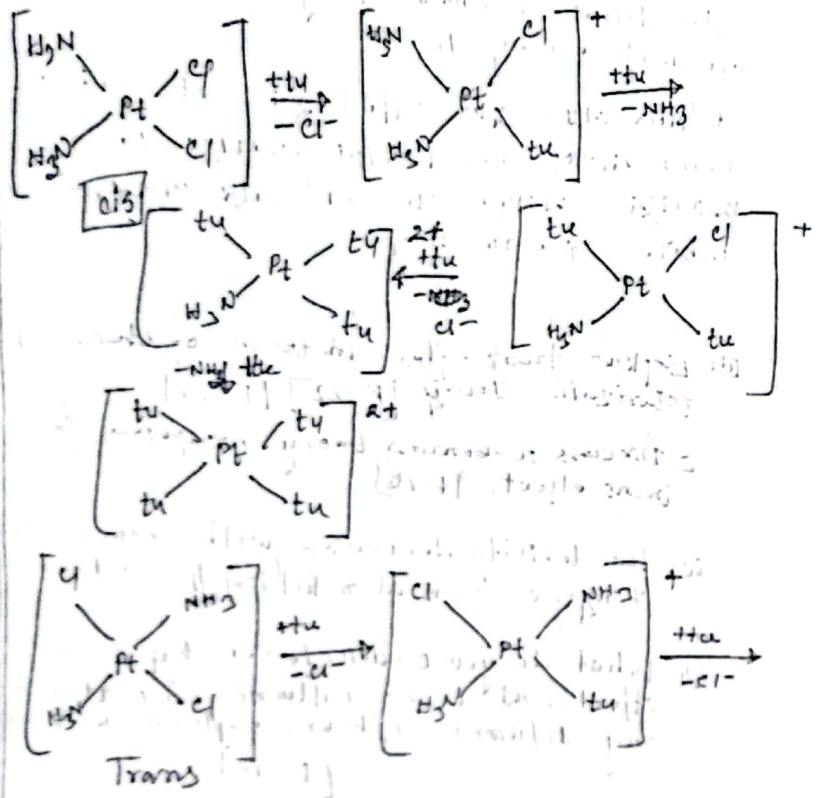
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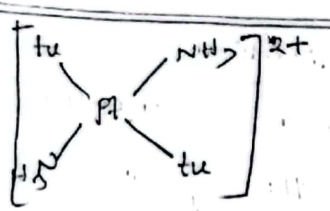
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A significant Application:

Differentiation of cis-trans isomers.

In cis isomers all the ligands are replaced by Thiourea (tu)





In trans isomer, only chloride/Halides are replaced by tu.

* NH_3 অণুর দ্বারা effect টুকু weak. এবং Pt-N bond সবচেয়ে inert তাই সবসময় trans অবস্থানে অবস্থিত NH_3 কে কখনো বিস্থাপিত করে না।

Explain trans effect in terms of electrostatic polarization theory. [F-22] [F-28]

Discuss π-bonding theory proposed for trans effect. [F-28]

The lability decreases with increase the charge on central metal ion. [F-19] [F-17]

What do you understand by trans effect and trans influence. Do they have any difference - if have, explain with example [F-17]

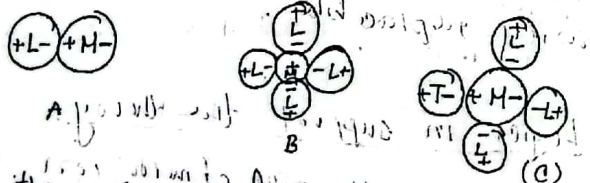
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* Explain reaction mechanism in trans and cis - $[M(LL)_2BX]$ (LL = bidentate such as en) [F-17]

* $[AlF_6]^{3-}$, $[SiF_6]^{2-}$, $[PF_6]^-$ and SF_6 are all six coordinate complexes.
Arrange them in order of decreasing ligand exchange rate and explain (2+2) [F-16]

* What is the difference between intermediate and transition state (F-16) [2]

Electrostatic polarization Energy



Due to the presence of ligand there created a dipole in metal ion which is drawn in the above figure

If there are four ligands around the central metal atom of the same types, then the resultant of ligands opposite to each other are cancell out. So, net dipole = 0.

Let, one of the 4 ligands is more polarizable in this case, there creates a dipole in the metal ion i.e. given in fig-(c). Here, same type of ligands effect is cancel out and no charges creat on those sites. So, In this situation, the trans position of 'T' has the ligand 'L' where the positive charge is less on the metal ion. As a result, M-L bond will be weak and distance increase. Hence, L is easily replaceable.

Logics in support the theory:

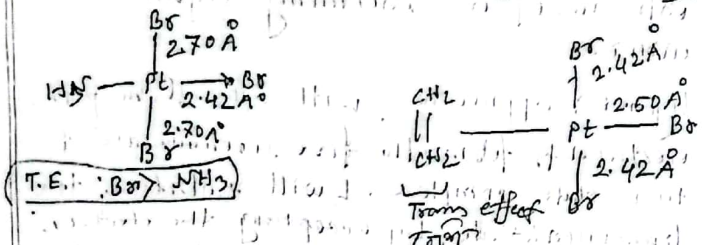
① The more the size of metal ion, the more the polarization will be. As a result trans effect also increase.

Pt (II) > Pd (II) > Ni (II)

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(2) Ligand's size, double or triple bond ↑ more polarizable and trans effect ↑ (বেশি হবে) Ex: $CN^-, CO, C_2H_4 > I^- > Br^- > Cl^-$

(3) $T-M-L$ (L, L). The trans position of T is M-L. This bond will be weaker and distance increases



π-bonding Theory:

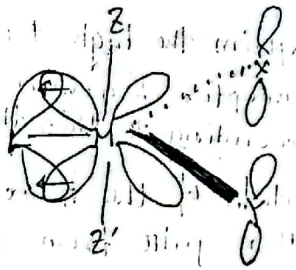
The theory explain the high trans effect shown by the π-acceptor ligands like CO, NO, CN^- C_2H_4 etc. According to this theory the vacant π or π* orbitals of the π-acceptor ligand accepts electron pair from the metal to form pi-bond between M-L which stabilizes the transition state.

The withdrawal of electron from metal by ligand increases the e-density toward M-L bond and diminishes in the direction of M-X.

L – M – X

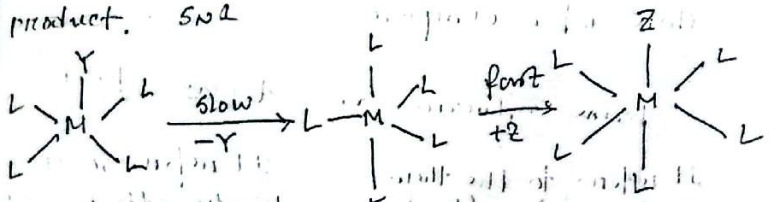
Thus the M-X bonds becomes weaker and entrance of an incoming ligand is become easier.

Ligand Y approaches with its lone pair of electron to form the five coordinated transition state complex. L will stabilize the transition state by accepting the electron density that the incoming ligand Y donates to the metal and therefore facilitates substitution.

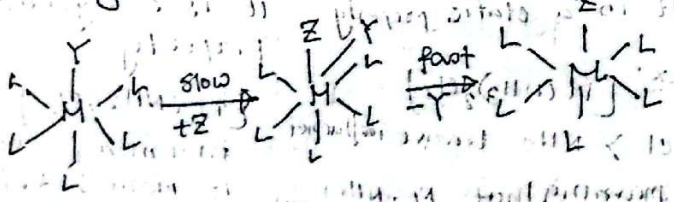


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In Dissociative Mechanism the metal ion dissociates the leaving ligand in the first step to form a five coordinate intermediate that is trigonal bipyramidal and then adds the incoming ligand in the second step to form the substituted product. SN1



In associative mechanism the metal ions first adds incoming ligand by accepting electron from the ligand and forms a seven coordinate compound and then dissociation occurs of the ligand. SN2



Trans effect : ट्रांस प्रभाव

Trans influence: Trans influence refers to the impact of a ligand on the length of the bond trans to it in the ground state of a complex.

Trans influence vs	trans effect.
It refers to the thermodynamic effect on the ground state bond length, meaning how much a ligand weakens the bond trans to it	It refers to the kinetic effect on the rate of substitution of a ligand, meaning how much a ligand facilitates the replacement of ligand trans to it during reaction.
It is a static property	it is a dynamic property.
Ex: $[Pt(NH_3)_2Cl_2]$ $Cl^- > NH_3$ trans influence meaning that Pt–NH_3 bond leng will be longer	$[Pt(NH_3)_2Cl_2]$ incoming ligand is more likely to replace the NH_3

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RAHM

ADVANCED CO-ORDINATION CHEMISTRY

Tutorial 2023

(1) Briefly discuss the factors that affect the crystal field splitting energy (4)

Sol^n:

CFSE (Δ_o) Depends on —

1) Nature of ligands: The more strong the ligand the more will be the splitting hence increase in Δ_o.

2) Oxidation state: Δ_o of first transition series is 4500 – 10500 cm^{-1} for M^{2+} and 14000 – 25000 cm^{-1} for M^{3+} ions. Increase of charge reduce the radius, reduces the M–L distance, thereby increase the field.

3) Atomic number of metal: Δ_o value for corresponding complex of metal ions in the same group and same oxidation state increases 30–50% on going from 1st to 2nd series and for next.

$[Co(NH_3)_6]^{3+}$ $\Delta_o = 23000\ cm^{-1}$ ($3d^6$)

$[Rh(NH_3)_6]^{3+}$ $\Delta_o = 34000\ cm^{-1}$ ($4d^6$)

$[Ir(NH_3)_6]^{3+}$ $\Delta_o = 41000\ cm^{-1}$ ($5d^6$)

M–L bonding is improved for more expanded 4d and 5d orbitals compared to compact 3d orbitals.

(2) Large Δ value. Explain —

Δ value depends on three factors which are already discussed.

(i) $[Cr(NH_3)_6]^{3+}$ and $[Cr(Cl_6)]^{3-}$

Here, NH_3 is stronger than Cl^-

But the oxidation state and metal series is same.

So, $[Cr(NH_3)_6]^{3+}$ has larger Δ value.

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(II) $[Co(H_2O)_6]^{3+}$ and $[Rh(H_2O)_6]^{3+}$

Here, both complex have same ligands and also o.s. of the metal. But Co is 3d series metal whereas Rh is 4d series metal so, $[Rh(H_2O)_6]$ has larger Δ_o value.

(III) $[Fe(H_2O)_6] < [Fe(CN)_6]^{4-}$

(IV) $[Co(NH_3)_6]^{3+} > [Cr Cl_6]^{3-}$

(V) $[Cr(H_2O)_6]^{3+} > [Cr(H_2O)_6]^{2+}$

(3) Explain the formation of the following on the basis of CFT

(i) Paramagnetic $[FeF_6]^{3-}$ (ii) diamag. $[Ni(CN)_4]^{2-}$

(iii) tetrahedral $[NiCl_4]^{2-}$

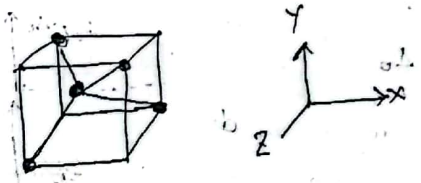
(4) What are the high spin and low spin complexes?

Soln: (3rd option) - (4 marks)

[5] Mode of splitting of d-orbitals in an octahedral field is just reverse of that in tetrahedral field - explain.

Soln:

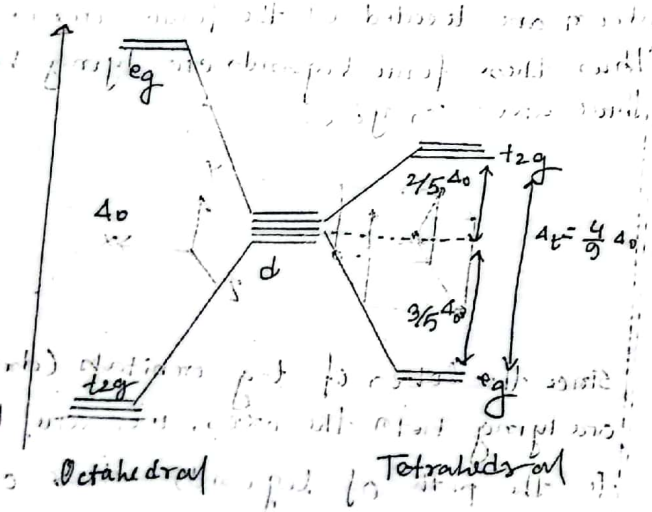
In tetrahedral complex, the four negative ligands placed at the four corners of the tetrahedron are located at the four corners of the cube. Thus these four ligands are lying between three axes x, y & z.



Since the lobes of t_{2g} orbitals (d_{xy}, d_{yz}, d_{zx}) are lying betn the axes, i.e, are lying closest to the path of ligands, these orbital will

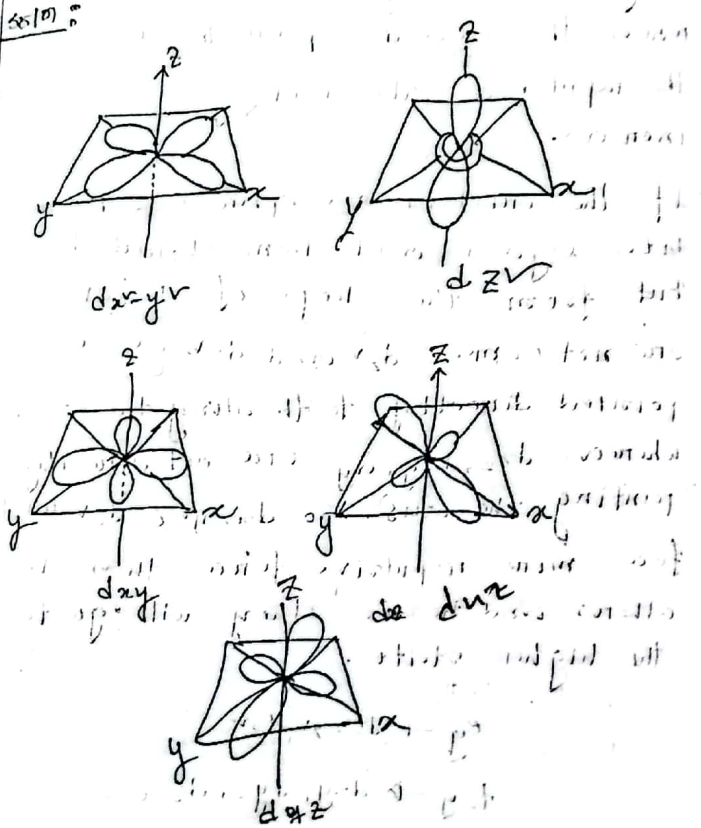
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experience greater repulsive force from the ligands than those of eg orbitals whose lobes are lying along axis. Thus the d orbitals split into two steps sets where t_{2g} has the higher energy. But in case of octahedral complex, ligands approaches to the metal along the axis. So, here eg will experience more repulsive force. So, the energy of eg will be higher.



5. (a) Draw the shapes of d and explain why they split into two groups t_{2g} and eg in an octahedral ligand field? discuss the factors of Δ. (5)

Ans:



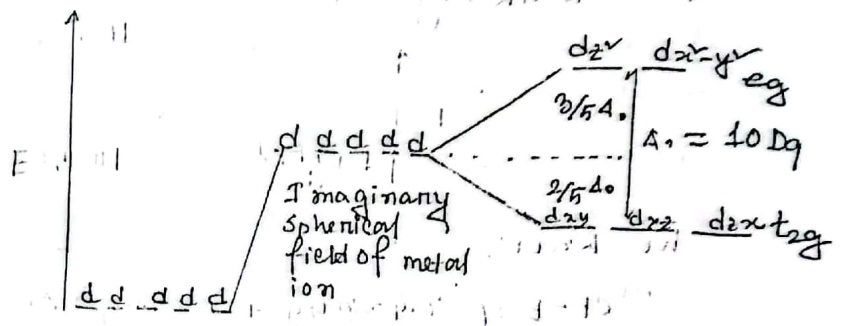
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In absence of ligands, any e⁻ can stay any of the five d-orbitals. Because, they all are degenerate orbitals. But when the ligand appears to it, due to the repulsion, the energy of d-orbitals increase.

If the orbitals are spherical, the distance betn ligands ~~and them should be~~ are equal But from the shape of d-orbitals are not same. d_{z^2} and $d_{x^2-y^2}$ are pointed directly ~~to it~~ along the axis whereas d_{xz}, d_{xy}, d_{yz} are not directly pointing the axis. So $d_{x^2-y^2}$ & d_{z^2} will face more repulsive force than the others and hence they will go to the higher state.

$e_g \rightarrow d_{x^2-y^2}, d_{z^2}$

$t_{2g} \rightarrow d_{xy}, d_{yz}, d_{zx}$.



*Factors discussed in previous.

(b) Which of the following pairs has a large Δ value? Explain your answer.

*Repeat Q

(c) Determine the no. of unpaired e⁻ and CFSE (ignore pairing energy):

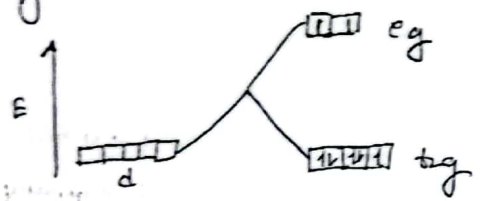
~~$[CoCl_4]^{2-}$~~, $[Zn(CN)_6]^{4-}$, $[Fe(H_2O)_6]^{3+}$

Co^{2+}: 3d ↑↓ ↑↓ ↑ ↑ ↑ 4s □

No. of unpaire e⁻ = 3

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With weak ligand,



We know,

$$CFSE = (-0.4 \times n_{t_{2g}} + 0.6 \times n_{e_g})\,\Delta_o$$

where,

$n_{t_{2g}}$ = no. of e⁻ in t_{2g}

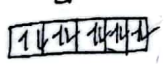
e_g = " " " " e_g

So, $CFSE = (-0.4 \times 5 + 0.6 \times 2)\,\Delta_o$

$= (-2 + 1.2)\,\Delta_o$

$= -0.8\,\Delta_o$

$[Zn(CN)_6]^{4-}$

Zn^{2+} : d^{10} 

no. of unpaired e⁻ = 0

d^0 $\quad n_{t_{2g}} = 6$

$n_{e_g} = 4$

$CFSE = (0.4 \times 6 + 0.6 \times 4)\,\Delta_o$

$= 0$

$[Fe(H_2O)_6]^{3+}$

Fe^{3+} : 3d  4s

No. of unpaired e⁻ = 5

d^5, H_2O is a weak field ligand.

So,

$n_{e_g} = 2$

$n_{t_{2g}} = 3$

$CFSE = (0.4 \times 3 + 0.6 \times 2)\,\Delta_o$

$= 0$

But, $[Fe(CN)_6]^{3-}$ CN^- strong ligand.



$CFSE = (0.4 \times 5) + 0$

$= -2.0\,\Delta_o$

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6 (a) What is Jahn-Teller theorem? On the basis of Jahn-Teller theorem explain the distortion of six coordinated complexes.

Soln:

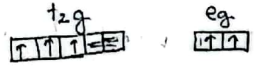
The Jahn-Teller theorem states, "any nonlinear molecular system in a degenerate electronic state will be unstable and will undergo distortion to form a system of lower symmetry and lower energy thereby removing the degeneracy."

For octahedral coordination,

If the electrons are equally placed in the d-orbitals, they exert same force of repulsion to the ligands. And hence uniform octahedral will form.

But molecules or complexes which have an unequally filled set of orbitals either t_{2g} or e_g will be distorted. In octahedral complexes the t_{2g} levels are in between the axis. So small distortion detected. Distortion resulting from uneven filling of the e_g orbitals are very important.

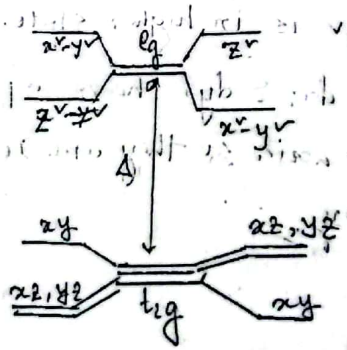
In this situation degeneracy removes energy difference is observed.

d^5 

symmetrical with weak ligand.

d^7 

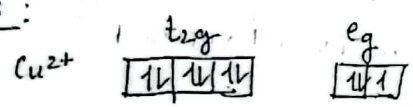
Unsymmetrical with strong ligand.



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(c) "In the crystal structure of CuF_2, the Cu^{2+} is coordinated by six F^- ; with four F^- at a distance of 1.93 Å and two F^- at a distance of 2.27 Å" — why

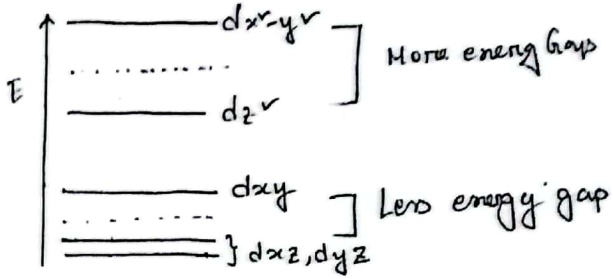
Ans:



We know that, F^- is a weak ligand. So, e⁻ will go to the e_g.

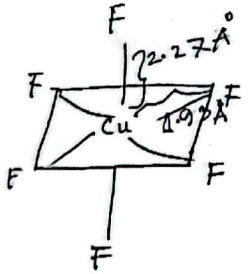
There are 4 ligands near to the $d_{x^2-y^2}$ & 2 ligand near to the d_{z^2}. So $d_{x^2-y^2}$ faces more repulsive force than d_{z^2}. So, $d_{x^2-y^2}$ is in higher state.

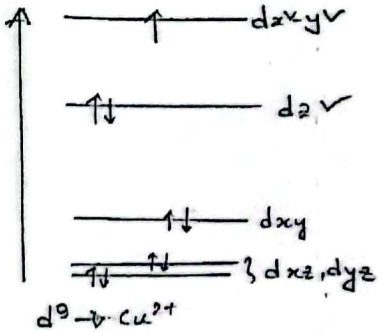
Again, d_{xz} & d_{yz} have a partial component to the z axis, so they are lower state than d_{xy}.



Unsymmetrically filled d-orditals energy.

Cu(II), the 9th e⁻ goes to the d_{z^2} conventionally. As a result, the 2 ligands situated along the z-axis experience more repulsive force and the distance will increase than the other four ligands distances.





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